

# The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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## SUMMARY

Voxel schemes for elastic scattering represent a heterogeneous medium as cubic cells, each carrying a single-site  $T$ -matrix and coupled through the background Green's tensor. They are built for media with no exact solution, where their error cannot be measured directly and a refinement study can converge smoothly to the wrong limit. A layer of identical voxels exercises such a scheme in full and has an exact answer. Because cubes tile space, it isolates the discretisation error exactly. For this layer we establish whether, how fast and with what error the scheme converges. (i) The cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling is *exactly* the continuum's, and the lattice kernel at normal incidence is in closed form. (ii) The single-site  $T$ -matrix follows from distributional moments of the Green's tensor over the cube. Its uniform-strain closure differs from a sphere's only by a cubic anisotropy of the shear response. Against exact scattering by a sphere it is exact statically, and it departs only by a real, closed-form  $(ka)^2$  term. (iii) The closure's self term cancels the kernel's, so the discrete system is a collocation of the continuum equation. (iv) It converges at second order, with leading error  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission. The opposite signs mean that no scalar correction of the single site can remove it. (v) A voxel

that also carries the first moment of its internal field converges at fourth order, with leading error  $-(kd)^4/720$ , for incident P, SV and SH at normal and oblique incidence. In a randomly stratified layer the orders are unchanged, and the first-moment voxel reaches an accuracy of  $10^{-6}$  with one voxel per model cell. Every result is derived symbolically and checked against an independent implementation or an exact solution.

**Key words:** Numerical modelling – Computational seismology – Wave scattering and diffraction – Theoretical seismology

## 1 INTRODUCTION

**The problem this paper serves.** Seismic waves in the crust are scattered by heterogeneity on every scale, and the models that describe it — tomographic images, full-waveform inversions, geological models — arrive as grids of cells of constant properties. A scattering calculation that uses such a model directly treats every cell as a scatterer: a voxel carrying a single-site  $T$ -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green’s tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the single voxel’s response is matched to cancel it, refinement converges to the wrong answer without revealing it.

**Why a homogeneous layer.** This paper therefore studies the one configuration in which a voxel scheme runs in full — every voxel, every coupling, the whole multiple-scattering series, at normal and at oblique incidence — and the exact answer is nevertheless known: a laterally uniform layer filled with identical voxels. Because cubes tile space, such a layer does not approximate a homogeneous layer; it *is* one. Any difference between its reflection and transmission and those of the continuous layer is therefore error of the discretisation alone, with no modelling error and no shape error behind which to hide. A sphere, the usual test object for volume-integral and discrete-dipole methods (Yurkin et al. 2006), cannot give this:

its error mixes the discretisation with the staircase approximation of a curved surface by cubes. The layer plays the part of the patch test in finite elements. A scheme that cannot reproduce a uniform medium correctly cannot be trusted in a heterogeneous one. What the layer reveals about the scheme's two ingredients — the coupling between voxels and the response of a single voxel — does not depend on the medium, because the same two ingredients are used wherever the scheme is applied.

**What the reader gains.** Four things that a heterogeneous test cannot provide. First, a proof, exact at normal incidence, that the voxel scheme converges to the continuum, and the specific ingredient — a coupling averaged over the source voxel — without which it converges to the wrong answer. Second, the leading error in closed form for every discretisation, so that a grid can be chosen for a required accuracy before the calculation is run. Third, a single-site  $T$ -matrix derived from first principles, with its error stated and checked against the exact solution for a sphere. Fourth, a voxel that converges at fourth order, also in a randomly stratified model (§9): for a reflection accuracy of  $10^{-6}$  it allows voxels 58 times larger in each direction than the standard one (§10).

**Why cubic voxels.** Cubes are the natural cell for three reasons. They *tile space*, so a uniform region discretised into cubes is still that uniform region, whereas spheres or point scatterers can only approximate it and carry a packing error of their own. Their *regular lattice* makes the interaction between cells translation invariant, so the coupling is a convolution — Toeplitz between planes, Bloch-periodic within them — and the whole system is solved by fast transforms. And a cube small compared with the wavelength has a *single-site response in closed form*, the elastic depolarisation of a cube (§6), so the only approximation a voxel introduces is the internal field it assumes.

**Multiple scattering on a lattice.** Replacing a heterogeneous body by a lattice of small scatterers that interact through the background Green's function is the multiple-scattering description of Foldy (1945) and Lax (1951). In electromagnetics its discretised form is the discrete dipole approximation (DDA; Purcell & Pennypacker 1973; Draine & Flatau 1994), whose accuracy has been studied in detail: each dipole's polarisability must account for the local field of the surrounding lattice, which the lattice dispersion relation fixes so that an infinite lattice propagates waves as the continuum does to a given order in  $kd$  (Draine & Goodman 1993); the error of the method converges at second order in the dipole spacing when shape errors are absent (Yurkin et al. 2006); and integrating the Green's tensor over each

cell rather than sampling it at the centre improves accuracy, most of all for strong contrasts (Chaumet et al. 2004). The elastic problem adds a structure the scalar and electromagnetic ones lack. A cubic voxel responds to the incident field with a force (through its density contrast) and a moment (through its stiffness contrast), so the state at a voxel has nine components — displacement and strain — and the single site’s self-interaction is the elastic depolarisation of a cube, the discrete counterpart of the inclusion tensor of Eshelby (1957).

**This paper.** The approach originates in the two-and-a-half-dimensional formulation of Nestor (1996): a medium divided into cells that interact through the background Green’s tensor, each carrying a single-site contrast renormalised by its own near field, with the multiple-scattering series summed. This paper extends it to cubic voxels. §2 sets out the discrete medium and §3 the continuous layer it must reproduce. §4 shows why the point coupling fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives the single-site  $T$ -matrix from moment integrals, tests it against the exact sphere, and shows that with it the discrete system is a collocation of the continuum equation. §7 gives that collocation’s second-order error in closed form, and §8 the fourth-order voxel, for incident P and S at normal and oblique incidence. §9 repeats the tests on a randomly stratified layer, and §10 discusses what the results imply for voxel schemes in general. Every result is derived symbolically in Mathematica and checked against an independent Python implementation or an exact solution.

## 2 SETTING

**The discrete medium.** Space is tiled by cubes of side  $d$  (half-side  $h = d/2$ ). Cube  $i$  carries a contrast  $\Delta\lambda, \Delta\mu, \Delta\rho$  from an isotropic background with Lamé parameters  $\lambda, \mu$ , density  $\rho$ , P and S speeds  $\alpha, \beta$  and P-wave modulus  $M_P = \lambda + 2\mu$ . Time dependence is  $e^{-i\omega t}$ ;  $k_P = \omega/\alpha$ ,  $k_S = \omega/\beta$ . Coordinates are ordered  $(z, x, y)$  with  $z$  down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the  $9 \times 9$  single-site  $T$ -matrix maps the exciting state to them, and the  $9 \times 9$  lattice Green’s tensor  $\mathbf{K}(i, j)$  maps a source voxel’s force and moment to the state at voxel  $i$ . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

with  $\psi^0$  the incident state. Strain rows of  $\mathbf{K}$  are receiver derivatives of the point Green's tensor  $\mathbf{G}$ ; moment columns are derivatives with respect to the receiver coordinate, symmetrised with a factor  $\frac{1}{2}$  on the shear components. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (3)$$

where  $\Delta \mathcal{C}$  is the  $6 \times 6$  contrast stiffness with  $\Delta \lambda + 2\Delta \mu$  and  $\Delta \lambda$  in the normal block and  $2\Delta \mu$  (not  $\Delta \mu$ ) on the shear diagonal.

**The laterally uniform layer.** A layer of thickness  $D$  is filled with  $n$  planes of voxels,  $d = D/n$ , and illuminated at normal incidence. The field is then laterally uniform, only the zero-lateral-wavenumber (specular) component of the lattice coupling acts, and Eq. (2) becomes exactly a chain of  $n$  planes coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

the sum running over the square lattice  $\mathbf{R}$  of pitch  $d$ ,  $m$  the plane separation in pitches. The question is whether, and how fast, this chain converges to the continuous layer as  $n \rightarrow \infty$  at fixed  $D$ .

**Implementations.** Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. “The package” below is the Python implementation of the voxel scheme (`cubic_scattering`): its cube  $T$ -matrix, its lattice kernels and its stratified propagator, which serve both as the object under test and, where validated, as a second implementation.

### 3 THE CONTINUOUS LAYER AND ITS THIN-LAYER EXPANSION

At normal incidence a plane force drives a one-dimensional problem per wave type, with modulus  $M$  ( $M_P$  for P,  $\mu$  for S), wavenumber  $k$  and plane Green's function  $g(z) = \frac{i}{2Mk} e^{ik|z|}$ . The layer's reflection and transmission follow from continuity of displacement and traction at its faces; they agree with the stratified propagator used as the reference in this work to  $2.5 \times 10^{-8}$ , P and S (`ContinuumLimit_Reference.wl`).

For a layer of thickness  $D$  with modulus  $M_1$  and density  $\rho_1$  in a background  $M_0, \rho_0$ , the reflection coefficient expands as

$$R = \frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[ \frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left( \frac{1}{M_1} - \frac{1}{M_0} \right) \right] + O(D^2), \quad (5)$$

exact in the contrast (`ContinuumLimit_ThinLayer.wl`): to first order a thin layer is an interface across which the traction jumps by the excess inertia and the displacement by the excess compliance. At oblique incidence the same statement holds with the first-order system  $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$  for the P–SV state  $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$ : the layer acts to  $O(D)$  as the jump  $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$ , which reproduces the layer’s  $O(D)$  reflection coefficients  $R_{PP}, R_{PS}$  to  $10^{-12}$ ; the series truncated at  $O(D^k)$  converges at exactly order  $k$ ; and a soft thin layer with shear modulus  $\mu_1 = D/\eta$  at fixed compliance  $\eta$  tends to the linear-slip interface as  $D \rightarrow 0$ . These expansions are the target a single plane of voxels must reproduce order by order.

#### 4 THE SPECULAR SUMS OF THE POINT GREEN’S TENSOR

By Poisson summation,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (6)$$

with  $\hat{\mathbf{G}}$  the two-dimensional transform of the Green’s tensor between planes. The  $\mathbf{g} = 0$  term is the continuum’s plane-wave term; the others are evanescent, decaying as  $e^{-2\pi m \sqrt{p^2 + q^2}}$  independently of  $d$ . In the strain-from-moment block they carry two powers of  $|\mathbf{g}| \propto 1/d$ , so their contribution per voxel volume does *not* vanish as  $d \rightarrow 0$ : at  $m = 1$  it is eleven times the continuum term (`ContinuumLimit_SpecularSums.wl`, matched to the package’s Ewald lattice sum to  $5 \times 10^{-13}$ ). This surviving term is static. Summed over all planes in slab order, the static lattice sum of the point kernel is exactly the depolarisation of a uniform plate —  $-1/M_P$  on  $\varepsilon_{zz} \leftarrow M_{zz}$  and  $-1/(2\mu)$  on the normal shears, per unit volume — plus a cubic-symmetric remainder (`ContinuumLimit_StaticLocal.wl`, to  $2 \times 10^{-10}$ ). The point kernel therefore reproduces the continuum only if that remainder is cancelled by the voxel’s own self term.

#### 5 THE TILING IDENTITY

Replace the point coupling by the *source-cell average*  $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$  — the field at a voxel centre of a uniform polarisation of the source voxel. Each Poisson order in Eq. (6) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (7)$$

which vanishes at every  $\mathbf{g} \neq 0$  of the reciprocal lattice since  $\text{sinc}(p\pi) = 0$  for integer  $p \neq 0$ . This is the Fourier form of the fact that cubes tile the plane,  $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$  for the cube's indicator  $\chi$ . Two exact statements follow.

**Between planes.** For  $m \neq 0$  every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (8)$$

the continuum's plane-wave term times the vertical form factor. The package's cell-averaged kernel agrees in all 81 entries to  $2.8 \times 10^{-16}$  (`ContinuumLimit_AveragedSums.wl`).

**Within a plane.** The sum over *all* cells of the plane, the voxel's own included, is the field at the mid-plane of a uniform plate of thickness  $d$ :

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (9)$$

where  $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$  is elementary: in the one-dimensional transform each entry is  $c_0 + A/(Mk_z^2 - \rho\omega^2)$ , so it averages to  $c_0/d + A(e^{ikh} - 1)/(dMk^2)$ . Its only nonzero entries are  $u \leftarrow f$  on the diagonal and the three normal strain-from-moment entries, whose local weights  $-1/M_P$  and  $-1/(2\mu)$  are the plate depolarisation of §4. Statically the identity holds for all 21 independent components of the strain-from-moment tensor to  $4 \times 10^{-9}$ , with every cell integral evaluated analytically (below).

**Consequence: an exact kernel with no lattice sum.** At normal incidence the same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (10)$$

with no Ewald splitting, near shell or asymptotic tail. The route it replaces — Ewald summation with a near shell averaged by quadrature and an  $O(d^2)$  Taylor tail beyond it — carries a static strain-from-moment error of  $9.4 \times 10^{-4}$  at its default settings, falling to  $2.6 \times 10^{-5}$  only when both the shell radius and the quadrature order are raised; Eq. (10) is now the package's kernel at normal incidence.

**Cell integrals without quadrature.** Write the static (Kelvin) tensor as  $G_{bc} = (c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$  with  $c_B = (1/\mu - 1/M_P)/8\pi$  and  $c_A = 1/(4\pi\mu) - c_B$ . Every cell integral of its derivatives is a cell integral of derivatives of  $1/r$  and  $r$ . One application of the divergence theorem moves it onto faces  $x_p = c_p \pm h$ , which never contain the origin, and there the two-dimensional antiderivative in the face coordinates is continuous, so the face integral is

a sum over the face's corners. The distributional term at  $r = 0$  is carried by the divergence theorem itself. (A triple antiderivative evaluated at the cube's corners is *not* usable for repeated derivatives: its arctangent terms jump across the coordinate planes through the origin, and differentiating twice there discards exactly that term.) The self term so obtained reproduces the classical closed form of the cube's depolarisation to machine precision.

## 6 THE SINGLE SITE: THE CUBE $T$ -MATRIX FROM MOMENT INTEGRALS

Every error of a voxel scheme is made at the single site (§10), so the single site is derived here rather than imported. The classical route imports it. For an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a uniform load uniform, with a depolarisation tensor in closed form (Eshelby 1957, 1959; Mura 1987); its dynamic generalisation is known for spheres and ellipsoids (Mikata & Nemat-Nasser 1990; Michelitsch et al. 2003). For a cube or a general polyhedron the static field of a uniform eigenstrain is also in closed form (Chiu 1977; Lee & Johnson 1978; Rodin 1996; Nozaki & Taya 2001), but the strain it produces is not uniform, so there is no exact uniform-strain inclusion to import. In scattering theory the same object appears as the self-consistent or effective-field closure of the volume integral equation (Mal & Knopoff 1967; Gubernatis et al. 1977; Kanaun & Levin 2008): a first-order (Born) scatterer (Wu & Aki 1985) renormalised by the geometric series of its own near field, the step that turns a contrast into an effective contrast. Nestor (1996) introduced the single site in this form, summing the series of the cell's self integral, with the singular near field evaluated over an infinitesimal cell. The moments below evaluate the same self integral over the finite cube, as a distribution. The elastic  $T$ -matrix of Waterman (1976) represents a scatterer in spherical multipoles, the natural basis for a sphere; a voxel in a lattice needs one compatible with the lattice, and the moments over the cube provide it. The construction below makes the approximation of a voxel explicit rather than implicit: it derives the cube's  $T$ -matrix from the moments of the elastodynamic Green's tensor over the cube, and identifies the uniform-strain closure as one truncation of an exact hierarchy.

### 6.1 The closed hierarchy

Let the voxel be  $V = [-h, h]^3$  with contrast  $\delta c_{ijkl} = \Delta\lambda\delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$  and  $\Delta\rho$ . Expand the internal displacement about the centre to second degree and project the Lippmann–Schwinger equation onto 1,  $x_p$  and  $x_px_q$ . The unknowns are  $u_i(0)$ ,  $\partial_p u_i(0)$  and

221  $\partial_p \partial_q u_i(0)$ , and the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta \rho G_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (11)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (12)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (13)$$

222 all fields evaluated at the centre. Its coefficients are six moments of the background Green's  
223 tensor  $\mathbf{G}$  over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \end{aligned} \quad (14)$$

224 with derivatives taken with respect to the source point,  $\partial' = -\partial$ . That sign reaches  $\mathbf{N}$  and  
225  $\mathbf{Q}$  only, and no symmetry, parity or scaling check can detect it, so it is fixed by convention.  
226 The moments are graded by  $(D, W)$ , the number of derivatives and the degree of the weight:  
227  $\mathbf{G}(0, 0)$ ,  $\mathbf{K}(0, 2)$ ,  $\mathbf{N}(1, 1)$ ,  $\mathbf{M}(2, 0)$ ,  $\mathbf{P}(2, 2)$ ,  $\mathbf{Q}(3, 1)$ .

## 228 6.2 The moments as distributions

229 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[ \delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (15)$$

230 (Kupradze 1963; Aki & Richards 2002). Expanding  $g_c$  in powers of  $r$  reduces every moment  
231 in Eq. (14) to a finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (16)$$

232 For  $m = -1$  and  $D \geq 2$  the classical integrand is not absolutely integrable.  $\partial^2(1/r)$  decays as  
233  $r^{-3}$ , its angular average vanishes, and the value of the integral depends on the shape of the  
234 region excluded around the origin. This is the source of the  $\delta$ -function term of the inclusion  
235 problem, which a numerical quadrature misses. Without it the depolarisation is wrong; for  
236 a sphere it vanishes altogether (§6.3). It is not added afterwards here. The derivatives are  
237 distributional from the start, and each is moved onto the voxel's faces by pairing with the  
238 test function  $w \mathbf{1}_V$ , using  $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$ :

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (17)$$

239 The surface term is evaluated on the faces, a distance  $h$  from the origin, where every derivative  
240 of  $r^m$  is smooth. The recursion lowers  $D$  and  $W$  together and ends on convergent integrals.

The face integrals are done in closed form by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p; ] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (18)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units  $(2 \pm \sqrt{3})^n$  of the Pell equation  $x^2 - 3y^2 = 1$ . Left unreduced, these cancel catastrophically: for example,  $\log(72010600134783751 - 41575339372323900\sqrt{3})$  is a cancellation of 35 digits. They are therefore reduced to  $n \log(2 - \sqrt{3})$ , which is exact. Because every power of  $r$  is paired this way, the dynamic moments keep the even-power terms of the expansion ( $k^2 r$ ,  $r^3$ , ...) as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of  $\mathbf{G}$  misses those terms, at a real  $O((kh)^2)$  cost (§6.4).

### 6.3 Parity, and the nine-component block

The cube is centrosymmetric. A moment that couples a field quantity of degree  $a$  to one of degree  $b$  therefore vanishes unless  $a + b$  is even.  $u$ ,  $\partial u$  and  $\partial\partial u$  have degrees 0, 1 and 2, so Eq. (12) contains the first gradient alone:  $\partial u$  is determined by its own block and couples to neither  $u$  nor  $\partial\partial u$ . The leading scattering by a stiffness contrast is the stress dipole  $\int_V \delta c : \nabla u dV$ , which needs only  $\partial u$ . The leading far field of the stiffness contrast is therefore fixed by Eq. (12), and *no treatment of the second-gradient block (13) can change it*. What Eq. (13) contributes reaches the far field only through the  $O((kh)^2)$  coupling in Eq. (11). That is the order, and the origin (the variation of the field across the voxel), of the collocation error of §7, which the first-moment voxel of §8 removes.

With  $\Delta\rho = 0$ , Eq. (12) is the  $9 \times 9$  operator  $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$ , built from  $\mathbf{M}$  alone. Under  $O_h$  the nine gradient components decompose as  $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$ , and the operator is diagonal in that basis. Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (19)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (20)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (21)$$

The density channel is the displacement block of Eq. (11) with the second gradients dropped, an amplification  $(1 - \omega^2 \Delta\rho \mathbf{G})^{-1}$  with  $\mathbf{G}$  the cube integral of  $\mathbf{G}$ .

**The cube against the sphere.** For a sphere, Eshelby's tensor gives a dilatational channel  $1 + \Delta K/(\lambda + 2\mu)$ , with  $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$ , and a single deviatoric channel  $1 + 2\Delta\mu S_{\text{sph}}$ ,

with  $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$ . The cube's dilatational channel, Eq. (19), is the sphere's exactly: its eigenvalue involves  $\mathbf{M}$  only through the contraction  $\int_V \partial_i \partial_n G_{in} dV$ , and the static Navier equation makes  $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$ , whose integral does not depend on the shape. The cube's shear channels differ from each other. Their isotropic average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (22)$$

The  $\sqrt{3}$  terms cancel in the average. What a cube adds to a sphere at this order is therefore a pure cubic anisotropy. It splits the deviatoric channel about the sphere's value, weighted 3 : 2 by the dimensions of  $T_{2g}$  and  $E_g$ , and the diagonal shear is the more depolarised. At the background of §7 ( $\nu = 7/32$ ),  $2\mu S_{\text{off}} = 0.4314$ ,  $2\mu S_{\text{diag}} = 0.5929$  and the sphere's value is 0.4960.

**Where Eshelby enters.** Equations (19)–(21) follow from the moment integrals and the group theory alone. No ellipsoidal result is used, and no  $\delta$  term is added by hand. What corresponds to Eshelby is the *truncation*. Keeping  $\partial u$  and discarding  $\partial \partial u$  is exactly the uniform-internal-strain ansatz, and by the uniformity theorem that ansatz is exact if and only if the inclusion is an ellipsoid. For a sphere the construction is therefore exact in the static limit, and for a cube it is an approximation. The hierarchy (11)–(13) states that approximation as the omitted blocks, rather than as an unquantified shape correction. A polynomial expansion about the centre cannot represent the edge and corner concentrations of an isolated cube's internal field at any finite degree. A voxel in a uniform layer has no such concentrations: its neighbours have the same properties, so its internal field is the continuum's, which is smooth. That is why a low-degree internal field suffices in a lattice (§§7–8) although it is only an approximation for an isolated cube.

**Against the exact sphere.** These statements are tested against the exact elastic Mie solution for a sphere, with no Eshelby formula in between. The concentration of Mie order  $n$  is  $E_n = a_n/a_n^{\text{Born}}$  (orders 0, 1, 2 are the dilatational, density and shear channels), extrapolated to  $ka \rightarrow 0$ . The formulation is applied to a ball of radius  $a$ , with its moment  $\mathbf{M}$  evaluated by the surface route of Eq. (17) on  $|\mathbf{r}| = a$ . Its dilatational and shear channels equal Mie's to  $2 \times 10^{-10}$ , and the density channel is 1. The cube's closed forms meet the same numbers. Its dilatational channel is Mie's  $E_0$  to  $2 \times 10^{-10}$ , and its isotropic shear average, Eq. (22), is the depolarisation read from Mie's  $E_2$  to  $10^{-8}$ . The same moment taken as a principal value,

over the ball with a small sphere excluded around the origin, has zero depolarisation in every channel. The whole static response of the sphere is the  $\delta$  term.

At finite frequency the closure uses the dynamic ball moments, which are in closed form. It departs from Mie at second order in  $ka$  and first order in the contrast, and the departure is closed-form too: the ball's moments are elementary, and Mie's low-frequency series with the contrast kept exact gives  $E_n$  to relative order  $(ka)^3$ . To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(kPa)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(kSa)^2 + (kPa)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(kSa)^2 + 2(kPa)^2\beta^2/\alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (23)$$

with no  $O(ka)$  term and no  $O((ka)^3)$  term. The imaginary  $(ka)^3$  terms, the radiation damping, agree exactly in all three channels. So the closure's only error at this order is the real  $(ka)^2$  term, the uniform-field ansatz's own. That is the order the hierarchy assigns to the omitted second gradients (§6.3), and it has the same origin, the variation of the field across the scatterer, as the collocation error of §7. Eq. (23) is derived symbolically (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from the numerical Mie solve and an independent implementation of the closure to  $5 \times 10^{-4}$ , which is the size of the next order in the contrast.

#### 6.4 The collocation closure

The collocation closure takes the internal field of a voxel as uniform, the truncation of §6.3 for the density and stiffness channels together. The voxel's own field at its centre is then  $\mathbf{S}_{\text{self}}\Delta e$  for internal state  $e$ , where  $\mathbf{S}_{\text{self}}$  is the nine-component form of the moments  $\mathbf{G}$  and  $\mathbf{M}$ . By parity it is block-diagonal. So  $e = \psi + \mathbf{S}_{\text{self}}\Delta e$  and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (24)$$

With the kernel (10) the self term cancels. Substituting  $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$  into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (25)$$

an identity checked on the matrices to  $4 \times 10^{-16}$ . Every block of  $\mathbf{K}_{\text{all}}$  is an *exact* cell integral of the one-dimensional continuum kernel (Eqs. (8), (9)), so Eq. (25) is precisely the collocation of the continuum Lippmann–Schwinger equation with a strain piecewise constant over voxels. It does not contain  $\mathbf{S}_{\text{self}}$  at all.

The package's cube  $T$ -matrix is this closure statically: both blocks agree with Eq. (24)

to  $2 \times 10^{-5}$  at  $k_S h = 0.01$ , the difference falling as  $(kh)^2$  (`ContinuumLimit_CubeT.w1`). It departs dynamically by about  $0.25(kh)^2$  through its far-field form-factor corrections. Its self integrals are exact in the static and radiation parts but omit the real even-power terms of the Green's tensor's expansion ( $k^2 r$ ,  $r^3$ , ...), an error of  $0.4\text{--}0.6(kh)^2$  against the analytic self term, whose series agrees with the closed-form elastodynamic tensor to  $10^{-14}$ . By Eq. (25) that error cannot enter the closure scheme.

## 7 CONVERGENCE, AND THE CLOSED-FORM ERROR

**Second order.** For a layer  $D = 2\text{ m}$  with the contrast  $\Delta\lambda = 2\text{ GPa}$ ,  $\Delta\mu = 1\text{ GPa}$ ,  $\Delta\rho = 100\text{ kg m}^{-3}$  in a background  $\alpha = 5000$ ,  $\beta = 3000\text{ m s}^{-1}$ ,  $\rho = 2500\text{ kg m}^{-3}$ , illuminated by a P plane force, the reflected displacement of the chain converges to the continuous layer's at order 2.000 for  $n = 4 \rightarrow 16$  at  $\omega = 60, 300$  and  $600\text{ rad s}^{-1}$ , with the collocation (25) and with the package *T*-matrix alike (`ContinuumLimit_Chain.w1`). The error depends on  $k_S h$  alone.

**The leading error in closed form.** In Eq. (25) the kernel is integrated exactly; the only approximation is the internal field's variation across a voxel. At first order in the contrast the discrete scattered field is therefore a midpoint rule applied to the continuum's Born integral, with the kernel's cell integral exact. For reflection the integrand carries the two-way phase  $e^{2ikz}$ ; for transmission it is flat. With  $k$  the background wavenumber of the wave concerned ( $k_P$  for an incident P wave,  $k_S$  for an incident S wave),  $R$  the reflection coefficient and  $T^{\text{sc}}$  the scattered part of the transmitted field, for *every*  $n$

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (26)$$

confirmed for  $n = 1$  to 16 to  $10^{-4}$  of the error itself (`ContinuumLimit_ErrorConstant.w1`); for an incident S wave, with  $k = k_S$ , the same forms and those of Eq. (29) hold to  $4 \times 10^{-4}$  (`ContinuumLimit_IncidentS.w1`, §8). At the full contrast the reflection constant is unchanged (0.124 against  $1/8$ ); the transmission constant grows to  $-0.092$ .

**Why a uniform-field single site stops at second order.** The two constants have opposite signs, so no scalar correction of  $\mathbf{T}$  — no single  $O((kh)^2)$  factor — can cancel both. The package *T*-matrix illustrates it: its form-factor corrections reduce the reflection error by a factor 4.45 and *increase* the transmission error by 1.7, at every  $n$ . The missing term in Eq. (26) is the first moment of the internal field over each voxel, which radiates through the derivative of the kernel. A fourth-order scheme must carry it: a single site with a gradient channel, and a kernel whose cell integrals include the kernel's derivative.

## 8 FOURTH ORDER: THE FIRST-MOMENT VOXEL

Let each voxel carry the mean *and* the first moment of its internal field. On voxel  $j$ , with local coordinate  $s \in [-h, h]$  and Legendre functions  $\phi_0 = 1$ ,  $\phi_1 = s/h$ , the field is  $w = \sum_b c_{b,j} \phi_b$ ,  $w = (u, \varepsilon)$  the displacement and strain of the P channel, and the polarisation is  $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$  with  $\Delta M = \Delta \lambda + 2\Delta \mu$ . Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z - z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (27)$$

with the same two functions on every voxel gives

$$\left( \int_i \phi_a^2 \right) c_{a,i} = \int_i \phi_a w^0 + \sum_{j,b} \left[ \int_i \int_j \phi_a(z) \mathbf{K}(z - z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (28)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$ . Between voxels the double integral is separable, a product of Legendre moments of  $e^{\pm i k z}$ ; within a voxel it is split at  $z' = z$  and the local term contributes  $-\int_i \phi_a \phi_b / M_P$ . Every entry is closed-form. Keeping only  $\phi_0$  recovers a second-order scheme; keeping  $\phi_0$  and  $\phi_1$  converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.w1`):

| $n$   | collocation                               | mean only                                 | mean and first moment                       |
|-------|-------------------------------------------|-------------------------------------------|---------------------------------------------|
| 1     | $1.8 \times 10^{-3} / 1.3 \times 10^{-3}$ | $1.2 \times 10^{-3} / 1.8 \times 10^{-3}$ | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   |
| 4     | $1.1 \times 10^{-4} / 8.3 \times 10^{-5}$ | $7.5 \times 10^{-5} / 1.1 \times 10^{-4}$ | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   |
| 16    | $7.0 \times 10^{-6} / 5.2 \times 10^{-6}$ | $4.7 \times 10^{-6} / 6.9 \times 10^{-6}$ | $4.2 \times 10^{-12} / 4.2 \times 10^{-12}$ |
| order | 2 / 2                                     | 2 / 2                                     | 4 / 4                                       |

(relative error of the scattered displacement, reflection / transmission,  $\omega = 300 \text{ rad s}^{-1}$ ; the order is 4.00 at  $\omega = 60$  and  $600 \text{ rad s}^{-1}$  too). A single plane of first-moment voxels represents the 2 m layer 6500 times more accurately than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full  $9 \times 9$  system to all printed digits.

**The leading error in closed form.** Without scattering, the Galerkin solution is the  $L^2$  projection  $\Pi$  of the incident field onto each voxel's basis, and the kernel is integrated over each voxel exactly; at first order in the contrast the error is therefore the kernel's cell integral applied to  $\Pi w^0 - w^0$ . The field varies as  $e^{i k s}$  across a voxel and the kernel toward an outside observer as  $e^{i q s}$ ,  $q = +k$  for reflection and  $q = -k$  for transmission, so for every scheme and

every  $n$

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (29)$$

with  $\Pi$  the centre value for collocation, the cell mean for the mean-only voxel and the projection onto  $\{1, s/h\}$  for the first-moment voxel (`ContinuumLimit_FourthOrderError.wl`):

|              | reflection                | transmission              |
|--------------|---------------------------|---------------------------|
| collocation  | $1 + (kd)^2/8 + O(d^4)$   | $1 - (kd)^2/24 + O(d^4)$  |
| mean only    | $1 + (kd)^2/12 + O(d^4)$  | $1 - (kd)^2/12 + O(d^4)$  |
| first moment | $1 - (kd)^4/720 + O(d^6)$ | $1 - (kd)^4/720 + O(d^6)$ |

The chain of Eq. (28) reproduces all six forms at  $n = 1$  to 8 to  $10^{-4}$  of the error, the remainder being the second order in the contrast. The first-moment voxel's leading error is the same in reflection and in transmission: its projection error is orthogonal to everything linear across the voxel, so the leading term no longer depends on the direction in which the field is observed.

**In three dimensions.** The three-dimensional first-moment voxel carries the nine-component state with Legendre weights  $\{1, x/h, y/h, z/h\}$  — 36 unknowns — tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (28) (`ContinuumLimit_Reduction3D.wl`). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and  $x/h, y/h$  are odd there. The mean and the  $z$ -moment carry weights that are constant laterally, so their lateral form factor  $\text{sinc}(g_x h) \text{sinc}(g_y h)$  vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in  $z$  alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor,  $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$ , does not vanish on the reciprocal lattice ( $F_1 = i(-1)^p/p\pi$ ), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift,  $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$ , so the tiling identity holds only to  $O(k_{\parallel} h)$ .

**Oblique incidence.** The scheme is built spectrally. By Poisson summation the Bloch coupling between planes  $i$  and  $j$  at lateral wavenumber  $\mathbf{k}_{\parallel}$  is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \int \int \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (30)$$

with  $L_a$  the lateral Legendre factors. The whole-space tensor decays as  $1/k_z^2$  and the nine-component state adds at most two powers of  $k_z$ , so each entry of  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$  is a  $\delta(z)$  of constant

weight plus, per mode,  $[i\alpha/2\gamma + i\beta \operatorname{sgn}(z)/2] e^{i\gamma|z|}$ ,  $\gamma^2 = k_W^2 - \kappa^2$ : every  $z$  integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at  $|p|, |q| \leq 8$ . Against the exact layer, for a P wave incident at  $20^\circ$  and  $40^\circ$  (`ContinuumLimit_Oblique.wl`):

|       | mean only            |                      | mean and first moments |                       |
|-------|----------------------|----------------------|------------------------|-----------------------|
| $n$   | $R_{PP}$             | $R_{PS}$             | $R_{PP}$               | $R_{PS}$              |
| 1     | $9.3 \times 10^{-4}$ | $1.7 \times 10^{-3}$ | $7.2 \times 10^{-8}$   | $4.1 \times 10^{-7}$  |
| 2     | $2.3 \times 10^{-4}$ | $4.2 \times 10^{-4}$ | $4.5 \times 10^{-9}$   | $2.5 \times 10^{-8}$  |
| 8     | $1.5 \times 10^{-5}$ | $2.7 \times 10^{-5}$ | $1.7 \times 10^{-11}$  | $9.9 \times 10^{-11}$ |
| order | 2                    | 2                    | 4                      | 4                     |

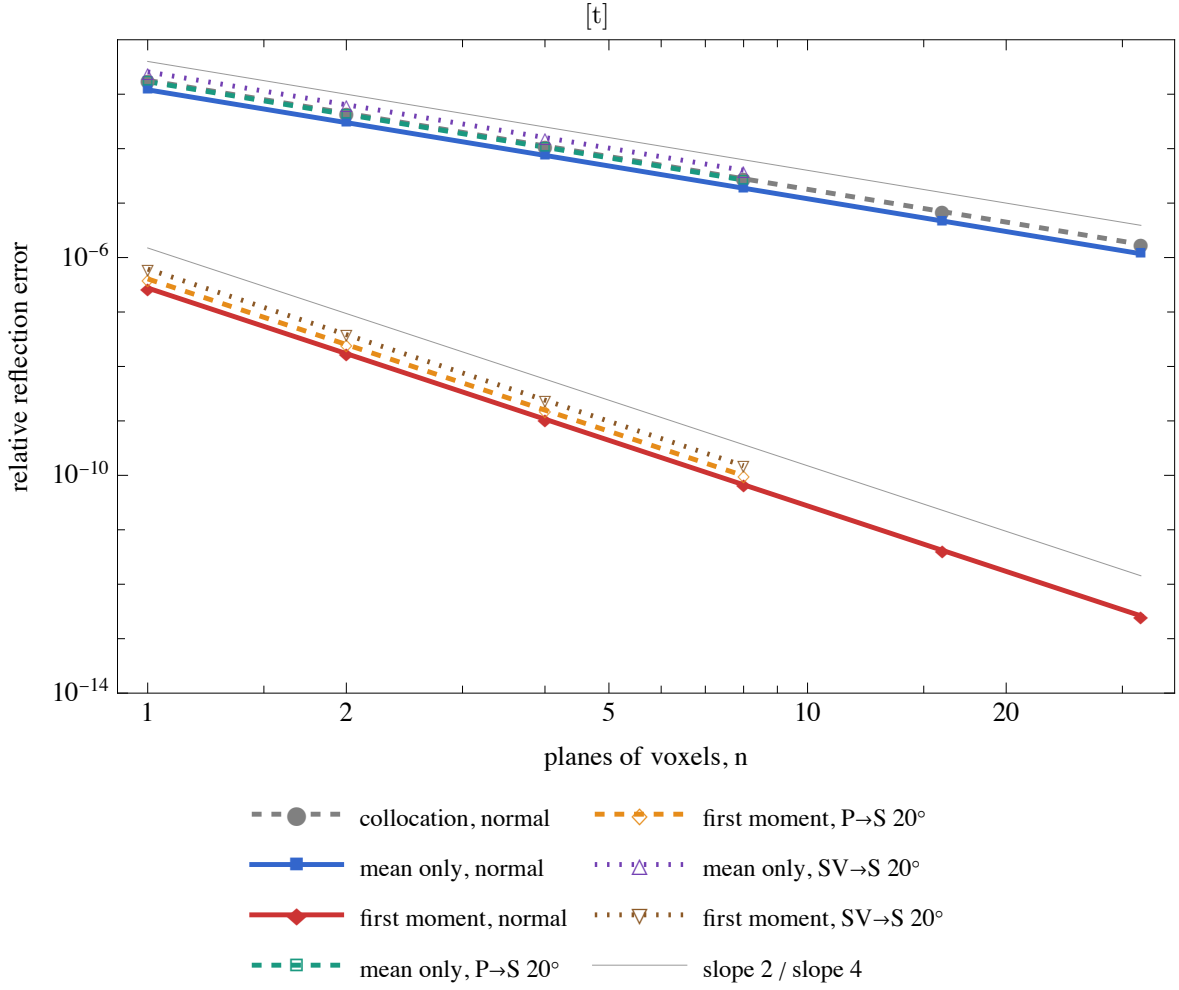
(relative error,  $20^\circ$ ; at  $40^\circ$  the orders are again 2 and 4). The mean-only voxel is second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at  $O(k_{\parallel}h)$  and the evanescent coupling of the gradient channels do not cost an order. At  $\mathbf{k}_{\parallel} = 0$  the construction reproduces Eq. (28) to all printed digits.

**Incident S.** Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (28) with  $M_P \rightarrow \mu$ ,  $k_P \rightarrow k_S$  and  $\Delta M \rightarrow \Delta\mu$ ; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full  $9 \times 9$  construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique incidence, against the exact layer (`ContinuumLimit_IncidentS.wl`):

|                                            | mean only   | mean and first moments |
|--------------------------------------------|-------------|------------------------|
| SV at $20^\circ$ : $R_{SS} / R_{SP}$       | 2.00 / 2.00 | 4.00 / 3.99            |
| SV at $30^\circ$ : $R_{SS} / R_{SP}$       | 1.98 / 2.00 | 3.97 / 4.00            |
| SH at $20^\circ$ and $40^\circ$ : $R_{SH}$ | 2.00        | 4.00                   |

(observed order,  $n = 2 \rightarrow 8$ ,  $\omega = 300 \text{ rad s}^{-1}$ ; the SV angles lie below the critical angle  $\arcsin(\beta/\alpha) = 36.9^\circ$ ). The orders, and with  $k = k_S$  the closed-form constants of Eqs. (26) and (29), are those of the incident P wave.

The spectral construction is implemented twice. The second implementation (`scripts/crosscheck_first_moment_voxel.py`) shares only the definition of the scheme: it takes the depth dependence of the transformed kernel from an independent residue construction, obtains the local weight as a numerical large- $k_z$  limit of the inverted Christoffel matrix, evaluates every double integral by quadrature and separates P from S by sampling the upgoing field at two



**Figure 1.** Relative reflection error of the discrete layer against the number of voxel planes  $n$  ( $D = 2$  m,  $\omega = 300$  rad s $^{-1}$ , the contrast of §7). Filled: normal incidence, P reflection — collocation (§6), the mean-only voxel and the first-moment voxel of Eq. (28). Dashed, open squares and diamonds: P incident at 20°, the converted S wave, mean-only (green) and first-moment (orange) voxels of Eq. (30). Dotted, open triangles: SV incident at 20°, the reflected S wave, mean-only (purple) and first-moment (brown). Grey: slopes 2 and 4.

depths. The raw reflected displacements of the two agree to  $9 \times 10^{-16}$  in all 36 cases compared — incident P, SV and SH, normal incidence and 20°,  $n = 1, 2, 4$ , with and without the first moments. Figure 1 collects the convergence.

## 9 A HETEROGENEOUS LAYER

A gridded Earth model is not a layer of identical cells. It is a set of model cells of constant properties with real interfaces between them. The last test takes such a model. A layer  $D =$

4m thick is divided into eight model cells of 0.5m, and in each one  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$  are drawn uniformly within  $\pm 15\%$  of the background (the values are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random stratification with physical contrasts. The voxels refine the model grid, with  $m$  planes of voxels in each model cell, so no voxel straddles a model interface; that is how a gridded model is discretised. At  $\omega = 1500 \text{ rad s}^{-1}$  the layer is  $k_P D = 1.2$  and  $k_S D = 2.0$  thick, so the field reverberates between the model interfaces. The exact answer is the product of the model cells' propagators. Its same-type reflection coefficients agree with those of the reflection–transmission recursion of Kennett (1983), computed independently, to  $10^{-14}$ . The schemes are those of §§7–8, with the contrast now changing from plane to plane; nothing else changes:

|                          |               | $m = 1$              | $m = 2$              | $m = 4$              | order |
|--------------------------|---------------|----------------------|----------------------|----------------------|-------|
| P, normal, $R$           | mean only     | $1.9 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $8.5 \times 10^{-7}$ | $5.3 \times 10^{-8}$ | $3.3 \times 10^{-9}$ | 4.00  |
| S, normal, $R$           | mean only     | $5.3 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $3.3 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $6.2 \times 10^{-6}$ | $3.9 \times 10^{-7}$ | $2.4 \times 10^{-8}$ | 4.00  |
| P, $20^\circ$ , $R_{PP}$ | mean only     | $1.4 \times 10^{-3}$ | $3.6 \times 10^{-4}$ | $9.0 \times 10^{-5}$ | 2.00  |
|                          | first moments | $3.3 \times 10^{-7}$ | $2.1 \times 10^{-8}$ | $1.3 \times 10^{-9}$ | 4.00  |
| P, $20^\circ$ , $R_{PS}$ | mean only     | $2.8 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | 2.00  |
|                          | first moments | $1.4 \times 10^{-6}$ | $8.8 \times 10^{-8}$ | $5.5 \times 10^{-9}$ | 4.00  |

(relative error against the exact stratified layer. At normal incidence the order is measured from  $m = 4$  to 16, where it is also 2.00 for collocation and 4.00 in transmission; at  $20^\circ$  it is measured from  $m = 2$  to 4.)

**The interfaces cost nothing.** Every order is the one measured for the uniform layer. At a model interface the strain of the true field jumps. A voxel basis that is discontinuous from voxel to voxel represents that jump exactly when the interface lies on a voxel boundary, so the interface adds no error of its own. The only requirement is that the voxels resolve the model grid. A voxel that straddled an interface would carry a contrast that is not uniform within it, a case not examined here.

**What fourth order buys.** For a reflection error of  $10^{-6}$  at normal incidence (P), the mean-only voxel needs 347 planes of voxels, about 43 per model cell. The first-moment voxel needs 8, one per model cell: the model grid itself, which already gives  $8.5 \times 10^{-7}$ . Counting unknowns (9 per voxel against 36), that is 3123 against 288 per vertical column of voxels, a factor of eleven. In three dimensions, with the lateral grid refined as the vertical one is, the

number of voxels falls by  $(347/8)^3 \approx 8 \times 10^4$  and the number of unknowns by about  $2 \times 10^4$ . The same model has been solved independently in Python, with a contrast for each plane: the raw reflected displacements agree with the Mathematica scheme to  $1.1 \times 10^{-15}$  at normal incidence and at  $20^\circ$ , with and without the first moments. Lateral heterogeneity needs the full three-dimensional solver; it is the first extension discussed below.

## 10 DISCUSSION

**Where the error of a voxel scheme lives.** The results separate the two approximations a voxel scheme makes. The *coupling* between voxels makes none: averaged over the source cell it is the continuum's cell integral exactly, between planes by the form factor's zeros and within a plane by the tiling identity, and the self term it omits is exactly the one the single site carries. Every error therefore comes from the *single site*, from the internal field it assumes. For a uniform field that error is the midpoint rule's, known in closed form; for a linear field it is the next term of the same expansion. The converse is the practical warning. A kernel that is not the cell integral — the point Green's tensor, or a cell average that does not match the single site's closure — leaves a local term that does not vanish as the voxels shrink (§4): the evanescent lattice orders carry a static contribution of fixed size per voxel volume. A scheme built that way converges to the wrong limit, and refinement cannot reveal it, because the error is scale-invariant.

**What the single site is.** The moment hierarchy of §6 turns the uniform-field single site from a modelling choice into a derived object with a stated error. Its static part is exact for a sphere, and for a cube it differs from the sphere's only by the cubic split of Eq. (22). Its dynamic part departs from exact scattering by a sphere only through the real  $(ka)^2$  term of Eq. (23); the radiation damping is exact. That term and the collocation error of Eq. (26) have the same origin, the variation of the field across the scatterer, and the same remedy, a richer internal field. In a lattice the remedy is cheap: the field a voxel must represent is the smooth continuum field, not the field of an isolated cube, with its concentrations at the edges and corners.

**Relation to the discrete dipole approximation.** The same partition appears in the electromagnetic literature as the choice of polarisability. The Clausius–Mossotti relation, the lattice dispersion relation (Draine & Goodman 1993) and the integrated Green tensor (Chaumet et al. 2004) all adjust the single site so that the lattice reproduces the continuum. For space-

filling cubes the adjustment is neither needed nor a free parameter: the tiling identity fixes the local field exactly, the self term of the collocation closure is the one that makes it cancel, and what is left is a closed-form error rather than an empirical one. The second-order convergence reported for the DDA by Yurkin et al. (2006) corresponds to the uniform-field voxel's, and the opposite-signed constants of Eq. (26) explain why a polarisability correction can improve one observable only at the expense of another, as the package's own form-factor corrections do in reflection and transmission. Going beyond second order requires a richer internal field, which is the point of higher-order bases for volume integral equations; the first-moment voxel is the lowest such basis, and in its spectral form the Galerkin double average makes the lattice sum converge without Ewald splitting.

**The cost of fourth order.** A first-moment voxel carries four times as many unknowns as a uniform one (36 against 9 in three dimensions). The closed forms show what that buys. To reach an error  $\varepsilon$  in reflection, the uniform voxel needs  $kd = \sqrt{8\varepsilon}$  and the first-moment voxel  $kd = (720\varepsilon)^{1/4}$ ; at  $\varepsilon = 10^{-6}$  that is  $kd = 2.8 \times 10^{-3}$  against 0.16, voxels 58 times larger in each direction. In a three-dimensional model the voxel count falls by  $58^3 \approx 2 \times 10^5$  and the number of unknowns by about  $5 \times 10^4$  (these are the Born-order constants; at the contrast studied the first-moment voxel's measured errors lie within 5% of them, while the uniform voxel's transmission constant grows with the contrast, so the comparison is if anything conservative). The stratified model of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-moment voxels for the same accuracy. The gain is largest precisely where it matters: when the grid of a geological model is coarse compared with the accuracy the wavefield must have.

**Limitations and outlook.** The results concern laterally uniform media — a homogeneous layer and a randomly stratified one — for which the discrete system is exactly a chain of planes; a laterally heterogeneous medium needs the first-moment voxel in a full three-dimensional solver, where the gradient channels couple through the evanescent lattice orders (§8) and the tiling identity holds only to  $O(k_{\parallel}h)$ . The numerical evidence is at moderate, physically typical contrasts (up to 15% in density and moduli) and at  $k_{\parallel}h \leq 0.3$ , the range in which the single-site  $T$ -matrix is validated; the closed forms are first order in the contrast, and the departures from them at the full contrast are of second order. What is established is the *order* of convergence and its leading constant, not the behaviour at strong contrast, where the single site's own closure may limit the accuracy before the lattice coupling does. The single site's dynamic error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§7–8). Two extensions follow directly. The first is the

three-dimensional solver itself. The second is a stratified *background* — a layered reference medium, as distinct from the stratified contrast of §9: the lattice coupling then carries the reverberation of the layering, which is not translation invariant in depth, and whether a tiling identity holds for it is the natural next question.

## 11 CONCLUSIONS

A lattice of cubic voxels reproduces a continuous elastic medium in a precise sense. Because cubes tile the plane, the source-cell-averaged coupling between voxels is the continuum’s cell integral exactly, and at normal incidence the lattice kernel is available in closed form with no lattice sum. The single site is derived rather than assumed. The moments of the Green’s tensor over the cube, taken as distributions, give a closed hierarchy whose first-gradient truncation is the uniform-strain closure. That closure is exact for a sphere in the static limit, and it departs from exact scattering by a sphere only by a real  $(ka)^2$  term, with the radiation damping exact. For a cube it differs from the sphere only by a cubic anisotropy of the shear response. The collocation closure carries the same self term as the lattice kernel, which cancels, so the discrete system is a collocation of the continuum equation. It converges at second order, with a leading error that is a midpoint-rule error of the internal field, known in closed form for every discretisation, and opposite in sign in reflection and transmission. A voxel that also carries the first moment of its internal field converges at fourth order — at normal incidence, where the three-dimensional scheme reduces exactly to one dimension, and at oblique incidence — with a leading error,  $-(kd)^4/720$ , that no longer depends on the direction of observation. The orders survive the interfaces of a randomly stratified model, provided the voxels resolve its grid, and there the first-moment voxel reaches an accuracy of  $10^{-6}$  with one voxel per model cell, where the uniform-field voxel needs about forty. The results, and the evidence for each, are:

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| Result                                                        | Evidence                                                                        |
|---------------------------------------------------------------|---------------------------------------------------------------------------------|
| Continuous layer, exact and to $O(D^k)$                       | reference to $2.5 \times 10^{-8}$ ; series order $k$                            |
| Point-kernel local term survives $d \rightarrow 0$            | $11 \times$ the continuum term at $m = 1$                                       |
| Averaged coupling between planes =<br>continuum $\times$ sinc | $2.8 \times 10^{-16}$ , 81 entries                                              |
| Tiling identity, Eq. (9)                                      | 21/21 components to $4 \times 10^{-9}$                                          |
| Exact kernel, Eq. (10)                                        | replaces a $9.4 \times 10^{-4}$ static error                                    |
| Moments as distributions, Eq. (17)                            | sum rule (18) exact; <b>M</b> , <b>Q</b> equal an independent<br>archive, 81/81 |
| Channels (19)–(21)                                            | a production code by another route, $2 \times 10^{-15}$                         |
| Cube vs sphere, Eq. (22)                                      | symbolic; and numerically at five $\nu$ , $4 \times 10^{-16}$                   |
| Ball closure = exact Mie, static                              | $2 \times 10^{-10}$ ; cube's bulk and shear average $10^{-8}$                   |
| 543 Ball closure vs Mie, dynamic, Eq. (23)                    | symbolic; numerical Mie $5 \times 10^{-4}$ ; $(ka)^3$ terms<br>exact            |
| Self term cancels, Eq. (25)                                   | $4 \times 10^{-16}$                                                             |
| Package $T$ = closure statically                              | $2 \times 10^{-5}$ at $k_S h = 0.01$                                            |
| Second-order convergence                                      | order 2.000, three frequencies                                                  |
| Closed-form error, Eq. (26)                                   | $n = 1$ –16, to $10^{-4}$ of the error                                          |
| First-moment voxel, Eq. (28)                                  | order 4.00, three frequencies                                                   |
| First-moment voxel, oblique, Eq. (30)                         | order 4, P and S, $20^\circ$ and $40^\circ$                                     |
| Incident SV and SH, all schemes                               | orders 2 and 4; constants with $k_S$ to $4 \times 10^{-4}$                      |
| First-moment error $-(kd)^4/720$ , Eq. (29)                   | $n = 1$ –8, to $10^{-4}$ of the error                                           |
| Independent implementation of Eq. (30)                        | $1.1 \times 10^{-15}$ , 40 cases: P, SV, SH, stratified                         |
| Random stratified layer, §9                                   | orders 2 and 4, P and S, normal and $20^\circ$                                  |
| Stratified reference against Kennett's recursion              | $10^{-14}$                                                                      |

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 546 Earth Sciences of the Australian National University, Canberra (Nestor 1996).

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## Data availability

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The Mathematica notebooks that derive and check every result (`Mathematica/`  
 549 `ContinuumLimit_*.wl`), the moment integrals of the single site (`Mathematica/`  
 550 `CubeMomentCore.wl`, `CubeA22Block.wl`, `CubeMomentArchiveCheck.wl`, `CubeShearSplit.`

wl, scripts/gate\_cube\_shear\_split.py and the test against exact Mie, scripts/gate\_sphere\_closure\_vs\_mie.py with Mathematica/SphereClosureDynamic.wl), the Python reference dumps and the independent implementation of the first-moment voxel (scripts/continuum\_limit\_\*.py, scripts/crosscheck\_first\_moment\_voxel.py), and the exact normal-incidence kernel (cubic\_scattering/cell\_averaged\_lattice.py) are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

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